

MATH1014 Calculus II Tutorial

Week 8

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1 Review

1.1 Type 1: Improper Integrals over Infinite Intervals

An integral is improper of Type 1 if the interval of integration is infinite. We define these integrals using limits:

1. $\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$
2. $\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$
3. $\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^\infty f(x) dx$ (for any real number c).

The improper integral is **convergent** if the corresponding limit exists and is finite. Otherwise, it is **divergent**. In case 3, *both* integrals must converge for the whole integral to be convergent.

1.2 Type 2: Improper Integrals with Discontinuous Integrands

An integral is improper of Type 2 if the integrand $f(x)$ has an infinite discontinuity (a vertical asymptote) at some point in the interval $[a, b]$.

1. If f is continuous on $[a, b)$ and discontinuous at b : $\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$
2. If f is continuous on $(a, b]$ and discontinuous at a : $\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$
3. If f has a discontinuity at c , where $a < c < b$: $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$

1.3 Comparison Theorem for Improper Integrals

Sometimes we only need to know whether an integral converges or diverges without calculating its exact value. Suppose f and g are continuous functions with $0 \leq f(x) \leq g(x)$ for $x \geq a$:

- If $\int_a^\infty g(x) dx$ is **convergent**, then $\int_a^\infty f(x) dx$ is also **convergent**.
- If $\int_a^\infty f(x) dx$ is **divergent**, then $\int_a^\infty g(x) dx$ is also **divergent**.

Useful Reference Integral (the p -integral): $\int_1^\infty \frac{1}{x^p} dx$ converges if $p > 1$ and diverges if $p \leq 1$.

Similarly, we often need comparison on a finite interval near a singular point. Suppose f and g are continuous functions on $(0, 1]$ and satisfy $0 \leq f(x) \leq g(x)$ for $0 < x \leq 1$:

- If $\int_0^1 g(x) dx$ is **convergent**, then $\int_0^1 f(x) dx$ is also **convergent**.
- If $\int_0^1 f(x) dx$ is **divergent**, then $\int_0^1 g(x) dx$ is also **divergent**.

Useful Reference Integral near 0 (the p -integral): $\int_0^1 \frac{1}{x^p} dx$ converges if $p < 1$ and diverges if $p \geq 1$.

2 Exercises

2.1 Evaluating Type 1 Improper Integrals

Exercise 2.1. Evaluate the following integral if it is convergent: $\int_0^{+\infty} e^{-2x} \sin(5x) dx$.

Solution. Idea. This is a Type 1 improper integral. We will evaluate the indefinite integral first using integration by parts, and then take the limit as the upper bound approaches infinity.

Let $I = \int e^{-2x} \sin(5x) dx$. We use integration by parts:

$$\int u dv = uv - \int v du$$

Let $u = e^{-2x}$ and $dv = \sin(5x) dx$. Then $du = -2e^{-2x} dx$ and $v = -\frac{1}{5} \cos(5x)$.

$$\begin{aligned} I &= -\frac{1}{5} e^{-2x} \cos(5x) - \int \left(-\frac{1}{5} \cos(5x) \right) (-2e^{-2x}) dx \\ &= -\frac{1}{5} e^{-2x} \cos(5x) - \frac{2}{5} \int e^{-2x} \cos(5x) dx \end{aligned}$$

Now apply integration by parts again to $\int e^{-2x} \cos(5x) dx$. Let $u = e^{-2x}$ and $dv = \cos(5x) dx$. Then $du = -2e^{-2x} dx$ and $v = \frac{1}{5} \sin(5x)$.

$$\begin{aligned} \int e^{-2x} \cos(5x) dx &= \frac{1}{5} e^{-2x} \sin(5x) - \int \frac{1}{5} \sin(5x) (-2e^{-2x}) dx \\ &= \frac{1}{5} e^{-2x} \sin(5x) + \frac{2}{5} \int e^{-2x} \sin(5x) dx \\ &= \frac{1}{5} e^{-2x} \sin(5x) + \frac{2}{5} I \end{aligned}$$

Substitute this back into our equation for I :

$$\begin{aligned} I &= -\frac{1}{5} e^{-2x} \cos(5x) - \frac{2}{5} \left(\frac{1}{5} e^{-2x} \sin(5x) + \frac{2}{5} I \right) \\ I &= -\frac{1}{5} e^{-2x} \cos(5x) - \frac{2}{25} e^{-2x} \sin(5x) - \frac{4}{25} I \end{aligned}$$

Add $\frac{4}{25}I$ to both sides:

$$\begin{aligned}\frac{29}{25}I &= e^{-2x} \left(-\frac{1}{5} \cos(5x) - \frac{2}{25} \sin(5x) \right) \\ I &= \frac{e^{-2x}}{29} (-5 \cos(5x) - 2 \sin(5x)) + C\end{aligned}$$

Now, we evaluate the improper integral:

$$\begin{aligned}\int_0^{+\infty} e^{-2x} \sin(5x) dx &= \lim_{t \rightarrow \infty} \left[\frac{e^{-2x}}{29} (-5 \cos(5x) - 2 \sin(5x)) \right]_0^t \\ &= \lim_{t \rightarrow \infty} \frac{e^{-2t}}{29} (-5 \cos(5t) - 2 \sin(5t)) - \frac{e^0}{29} (-5 \cos(0) - 2 \sin(0))\end{aligned}$$

Since $\cos(5t)$ and $\sin(5t)$ are bounded between -1 and 1 , and $\lim_{t \rightarrow \infty} e^{-2t} = 0$, the limit of the first term is 0 by the Squeeze Theorem. Thus, the integral evaluates to:

$$0 - \frac{1}{29}(-5) = \frac{5}{29}$$

The integral converges to $\frac{5}{29}$. □

Exercise 2.2. Evaluate the following integral if it is convergent: $\int_{-\infty}^{+\infty} \frac{1}{x^2 + x + 1} dx$.

Solution. Idea. We complete the square in the denominator to use the arctangent integration formula.

First, complete the square for $x^2 + x + 1$:

$$x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 - \frac{1}{4} + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} = \left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2$$

The indefinite integral is:

$$\int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx = \frac{1}{\frac{\sqrt{3}}{2}} \arctan\left(\frac{x + \frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + C = \frac{2}{\sqrt{3}} \arctan\left(\frac{2x + 1}{\sqrt{3}}\right) + C$$

Since the interval is $(-\infty, \infty)$, we formally split the integral at $x = 0$ (or just evaluate the limits at both infinities):

$$\begin{aligned}\int_{-\infty}^{+\infty} \frac{1}{x^2 + x + 1} dx &= \lim_{t \rightarrow \infty} \int_0^t \frac{1}{x^2 + x + 1} dx + \lim_{s \rightarrow -\infty} \int_s^0 \frac{1}{x^2 + x + 1} dx \\ &= \lim_{t \rightarrow \infty} \left[\frac{2}{\sqrt{3}} \arctan\left(\frac{2x + 1}{\sqrt{3}}\right) \right]_0^t + \lim_{s \rightarrow -\infty} \left[\frac{2}{\sqrt{3}} \arctan\left(\frac{2x + 1}{\sqrt{3}}\right) \right]_s^0\end{aligned}$$

Notice that the terms evaluated at 0 will cancel each other out, so we only need the limits:

$$\begin{aligned}&= \lim_{t \rightarrow \infty} \frac{2}{\sqrt{3}} \arctan\left(\frac{2t + 1}{\sqrt{3}}\right) - \lim_{s \rightarrow -\infty} \frac{2}{\sqrt{3}} \arctan\left(\frac{2s + 1}{\sqrt{3}}\right) \\ &= \frac{2}{\sqrt{3}} \left(\frac{\pi}{2}\right) - \frac{2}{\sqrt{3}} \left(-\frac{\pi}{2}\right) \\ &= \frac{2\pi}{\sqrt{3}}\end{aligned}$$

The integral converges to $\frac{2\pi}{\sqrt{3}}$. □

Exercise 2.3. Evaluate the following integral if it is convergent: $\int_0^{+\infty} x e^{ax^2} dx$ (discuss the value of constant a).

Solution. Idea. The convergence depends entirely on the sign of a . We discuss three cases: $a = 0$, $a > 0$, and $a < 0$.

Case 1: $a = 0$

The integral becomes $\int_0^{+\infty} x dx$.

$$\lim_{t \rightarrow \infty} \int_0^t x dx = \lim_{t \rightarrow \infty} \left[\frac{x^2}{2} \right]_0^t = \lim_{t \rightarrow \infty} \frac{t^2}{2} = \infty$$

The integral is divergent.

Case 2: $a \neq 0$

We use substitution $u = ax^2$, $du = 2ax dx \implies x dx = \frac{1}{2a} du$. The indefinite integral is:

$$\int x e^{ax^2} dx = \int \frac{1}{2a} e^u du = \frac{1}{2a} e^u = \frac{1}{2a} e^{ax^2} + C$$

Now we evaluate the improper integral:

$$\int_0^{+\infty} x e^{ax^2} dx = \lim_{t \rightarrow \infty} \left[\frac{1}{2a} e^{ax^2} \right]_0^t = \lim_{t \rightarrow \infty} \left(\frac{1}{2a} e^{at^2} - \frac{1}{2a} \right)$$

- **If $a > 0$:** As $t \rightarrow \infty$, $e^{at^2} \rightarrow \infty$. Thus, the limit does not exist (it goes to infinity), and the integral is **divergent**.
- **If $a < 0$:** As $t \rightarrow \infty$, $e^{at^2} \rightarrow 0$. Thus, the limit is $0 - \frac{1}{2a} = -\frac{1}{2a}$. The integral is **convergent**.

In conclusion, the integral converges to $-\frac{1}{2a}$ if $a < 0$, and diverges if $a \geq 0$. □

2.2 Evaluating Type 2 Improper Integrals

Exercise 2.4. Evaluate the following integral if it is convergent: $\int_0^1 \frac{1}{(2-x)\sqrt{1-x}} dx$.

Solution. Idea. The integrand has a vertical asymptote at $x = 1$, making it a Type 2 improper integral. We will evaluate it using limits and the substitution $u = \sqrt{1-x}$.

By definition, we write:

$$\int_0^1 \frac{1}{(2-x)\sqrt{1-x}} dx = \lim_{c \rightarrow 1^-} \int_0^c \frac{1}{(2-x)\sqrt{1-x}} dx$$

Let $u = \sqrt{1-x}$. Then $u^2 = 1-x$, so $x = 1-u^2$, and $dx = -2u du$. Also, $2-x = 2-(1-u^2) = 1+u^2$.

When $x = 0$, $u = \sqrt{1-0} = 1$.

When $x = c$, $u = \sqrt{1-c}$.

$$\begin{aligned}\int_0^c \frac{1}{(2-x)\sqrt{1-x}} dx &= \int_1^{\sqrt{1-c}} \frac{1}{(1+u^2)u} (-2u) du \\&= \int_1^{\sqrt{1-c}} \frac{-2}{1+u^2} du \\&= \int_{\sqrt{1-c}}^1 \frac{2}{1+u^2} du \\&= [2 \arctan u]_{\sqrt{1-c}}^1 \\&= 2 \arctan(1) - 2 \arctan(\sqrt{1-c}) \\&= 2 \left(\frac{\pi}{4} \right) - 2 \arctan(\sqrt{1-c}) = \frac{\pi}{2} - 2 \arctan(\sqrt{1-c})\end{aligned}$$

Now, taking the limit as $c \rightarrow 1^-$:

$$\lim_{c \rightarrow 1^-} \left(\frac{\pi}{2} - 2 \arctan(\sqrt{1-c}) \right) = \frac{\pi}{2} - 2 \arctan(0) = \frac{\pi}{2}$$

The integral converges to $\frac{\pi}{2}$. □

Exercise 2.5. Evaluate the following integral if it is convergent: $\int_{-1}^1 \frac{1}{x^3} \sin\left(\frac{1}{x^2}\right) dx$.

Solution. Idea. The integrand is discontinuous at $x = 0$, which is inside the interval $[-1, 1]$. We must split the integral at $x = 0$.

$$\int_{-1}^1 \frac{1}{x^3} \sin\left(\frac{1}{x^2}\right) dx = \int_{-1}^0 \frac{1}{x^3} \sin\left(\frac{1}{x^2}\right) dx + \int_0^1 \frac{1}{x^3} \sin\left(\frac{1}{x^2}\right) dx$$

If either of these integrals diverges, the whole integral diverges. Let's analyze the second one:

$$\int_0^1 \frac{1}{x^3} \sin\left(\frac{1}{x^2}\right) dx = \lim_{c \rightarrow 0^+} \int_c^1 \frac{1}{x^3} \sin\left(\frac{1}{x^2}\right) dx$$

Let $u = \frac{1}{x^2}$, then $du = -\frac{2}{x^3} dx$, so $\frac{1}{x^3} dx = -\frac{1}{2} du$. When $x = c$, $u = \frac{1}{c^2}$. When $x = 1$, $u = 1$.

$$\begin{aligned}\int_c^1 \frac{1}{x^3} \sin\left(\frac{1}{x^2}\right) dx &= \int_{1/c^2}^1 \sin(u) \left(-\frac{1}{2}\right) du \\&= \frac{1}{2} \int_1^{1/c^2} \sin(u) du \\&= \frac{1}{2} [-\cos u]_1^{1/c^2} \\&= \frac{1}{2} \cos(1) - \frac{1}{2} \cos\left(\frac{1}{c^2}\right)\end{aligned}$$

Now, we must evaluate the limit:

$$\lim_{c \rightarrow 0^+} \left(\frac{1}{2} \cos(1) - \frac{1}{2} \cos\left(\frac{1}{c^2}\right) \right)$$

As $c \rightarrow 0^+$, the term $\frac{1}{c^2} \rightarrow \infty$. The function $\cos(x)$ oscillates between -1 and 1 as its argument goes to infinity, meaning the limit $\lim_{c \rightarrow 0^+} \cos\left(\frac{1}{c^2}\right)$ **does not exist**.

Because this limit does not exist, the integral $\int_0^1$ diverges. Consequently, the original integral $\int_{-1}^1$ is **divergent**. □

2.3 Convergence Tests for Improper Integrals

Exercise 2.6. Check whether the following integrals converge or not.

1. $\int_1^{+\infty} \frac{1}{\sqrt{x^3 + \ln x + 1}} dx$
2. $\int_1^{+\infty} \frac{\arctan x}{1 + x^3} dx$
3. $\int_0^{+\infty} \frac{1}{1 + x |\sin x|} dx$

Solution. Idea. For these problems, we do not need to evaluate the integral exactly. We will use the Comparison Theorem by finding a simpler function $g(x)$ to compare with.

(1) Consider $f(x) = \frac{1}{\sqrt{x^3 + \ln x + 1}}$ for $x \geq 1$.

Since $x \geq 1$, we know that $\ln x \geq 0$. Therefore:

$$x^3 + \ln x + 1 > x^3$$

Taking the square root and then the reciprocal of both sides (which reverses the inequality):

$$\frac{1}{\sqrt{x^3 + \ln x + 1}} < \frac{1}{\sqrt{x^3}} = \frac{1}{x^{3/2}}$$

We compare our integral to $\int_1^{+\infty} \frac{1}{x^{3/2}} dx$. This is a p -integral with $p = \frac{3}{2} > 1$, so it **converges**. Since $0 < f(x) < \frac{1}{x^{3/2}}$, by the Comparison Theorem, the original integral **converges**.

(2) Consider $f(x) = \frac{\arctan x}{1 + x^3}$ for $x \geq 1$.

We know that for $x \geq 1$, $0 < \arctan x < \frac{\pi}{2}$. Also, $1 + x^3 > x^3$, which implies $\frac{1}{1 + x^3} < \frac{1}{x^3}$. Combining these two inequalities, we get:

$$0 < \frac{\arctan x}{1 + x^3} < \frac{\pi/2}{x^3}$$

The integral $\int_1^{+\infty} \frac{\pi/2}{x^3} dx = \frac{\pi}{2} \int_1^{+\infty} \frac{1}{x^3} dx$. This is a p -integral with $p = 3 > 1$, so it **converges**. By the Comparison Theorem, the original integral **converges**.

(3) Consider $f(x) = \frac{1}{1 + x |\sin x|}$ for $x \geq 0$.

We know that $0 \leq |\sin x| \leq 1$. Multiplying by x (since $x \geq 0$), we get:

$$0 \leq x |\sin x| \leq x$$

Adding 1 to all sides:

$$1 \leq 1 + x |\sin x| \leq 1 + x$$

Taking the reciprocal reverses the inequality:

$$\frac{1}{1 + x |\sin x|} \geq \frac{1}{1 + x} > 0$$

Let's evaluate the integral of the smaller function:

$$\int_0^{+\infty} \frac{1}{1 + x} dx = \lim_{t \rightarrow \infty} [\ln |1 + x|]_0^t = \lim_{t \rightarrow \infty} \ln(1 + t) = \infty$$

Since the smaller integral diverges to infinity, our original integral is larger than infinity. By the Comparison Theorem, the original integral **diverges**. $\square$

2.4 Advanced Substitution in Improper Integrals

Exercise 2.7. Suppose $a > 0$, evaluate:

$$\int_0^{+\infty} f\left(\frac{x}{a} + \frac{a}{x}\right) \frac{\ln x}{x} dx - \ln a \int_0^{+\infty} f\left(\frac{x}{a} + \frac{a}{x}\right) \frac{1}{x} dx$$

Solution. Idea. Combine the two integrals into one, split the integration interval at $x = a$, and use a smart substitution $x = \frac{a^2}{t}$ on the $[a, +\infty)$ part to show it cancels out the $(0, a]$ part.

First, combine the two integrals:

$$I = \int_0^{+\infty} f\left(\frac{x}{a} + \frac{a}{x}\right) \frac{\ln x - \ln a}{x} dx$$

We split the interval $(0, +\infty)$ into two parts: $(0, a]$ and $[a, +\infty)$:

$$I = \int_0^a f\left(\frac{x}{a} + \frac{a}{x}\right) \frac{\ln x - \ln a}{x} dx + \int_a^{+\infty} f\left(\frac{x}{a} + \frac{a}{x}\right) \frac{\ln x - \ln a}{x} dx = I_1 + I_2$$

Let's focus on $I_2 = \int_a^{+\infty} f\left(\frac{x}{a} + \frac{a}{x}\right) \frac{\ln x - \ln a}{x} dx$.

We apply the substitution $x = \frac{a^2}{t}$. Then $dx = -\frac{a^2}{t^2} dt$. Let's see how the boundaries and the integrand terms change:

- When $x \rightarrow +\infty$, $t \rightarrow 0^+$.
- When $x = a$, $t = \frac{a^2}{a} = a$.
- The argument of f : $\frac{x}{a} + \frac{a}{x} = \frac{a^2/t}{a} + \frac{a}{a^2/t} = \frac{a}{t} + \frac{t}{a}$. This is completely symmetric!
- The logarithmic part: $\ln x - \ln a = \ln\left(\frac{a^2}{t}\right) - \ln a = \ln(a^2) - \ln t - \ln a = 2 \ln a - \ln t - \ln a = \ln a - \ln t = -(\ln t - \ln a)$.
- The denominator $x = \frac{a^2}{t}$.

Substituting these into I_2 :

$$\begin{aligned} I_2 &= \int_a^0 f\left(\frac{a}{t} + \frac{t}{a}\right) \frac{-(\ln t - \ln a)}{a^2/t} \left(-\frac{a^2}{t^2}\right) dt \\ &= \int_a^0 f\left(\frac{t}{a} + \frac{a}{t}\right) \frac{\ln t - \ln a}{t} dt \end{aligned}$$

Flipping the bounds of integration from $\int_a^0$ to $\int_0^a$ introduces a negative sign:

$$I_2 = - \int_0^a f\left(\frac{t}{a} + \frac{a}{t}\right) \frac{\ln t - \ln a}{t} dt$$

Notice that t is just a dummy variable of integration. We can replace t with x , and we get exactly the negative of I_1 :

$$I_2 = - \int_0^a f\left(\frac{x}{a} + \frac{a}{x}\right) \frac{\ln x - \ln a}{x} dx = -I_1$$

Therefore, the original combined integral is:

$$I = I_1 + I_2 = I_1 - I_1 = 0$$

The final value of the expression is 0. □